K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

COMPUTER SCIENCE FUNDAMENTALS AND CAREER PATHWAYS

B.TECH CSE CORE SECTION (A) SEMESTER - 1

COURSE CODE : ETCCCP105

ASSIGNMENT TITLE :

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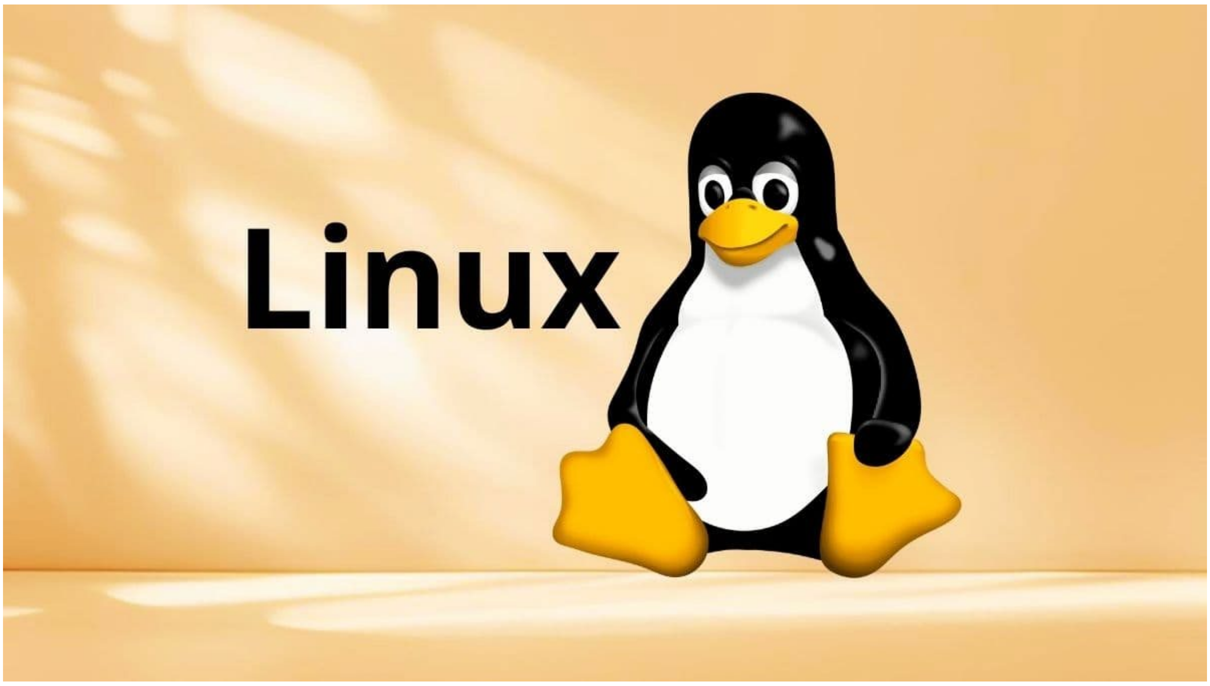
ROLL NUMBER : 2501010070

SUBMITTED TO : RAJESH KUMAR SIR

DATE OF SUBMISSION : 10 NOVEMBER 2025

INTRODUCTION



Linux is a free and open-source operating system kernel that was first released by Linus Torvalds in 1991.

It is a versatile and widely adopted operating system that forms the backbone of many academic, industrial, and technological environments. Its robustness, security features, and extensive scripting capabilities make it an essential tool for computer science and IT professionals.

This assignment aims to provide practical, hands-on experience with Linux installation, shell commands, and shell scripting to automate routine tasks.

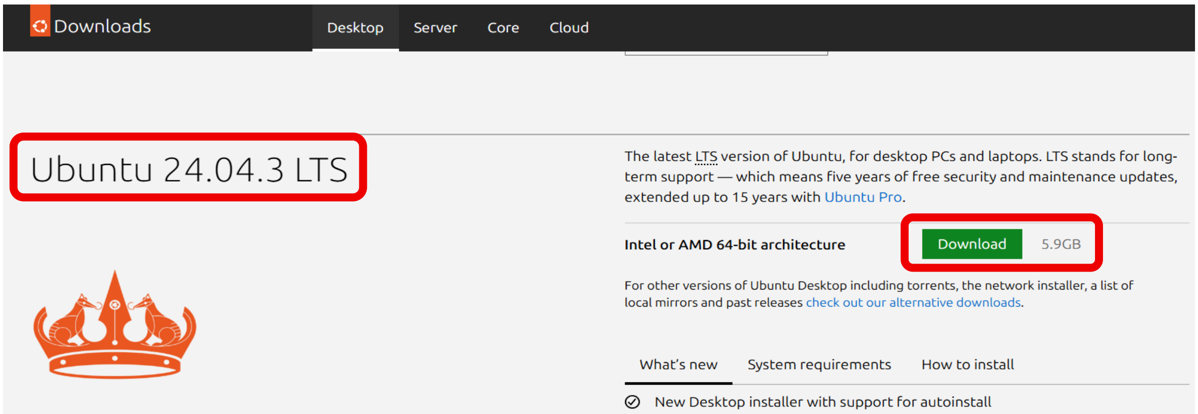
In this assignment, I learned how to write and run simple shell scripts on Linux. The tasks involved making scripts that help with everyday technical work, like backing up files, monitoring CPU and memory usage, and downloading files from the internet. This work helped me to understand basic Linux commands and how to automate different jobs using a shell script.

LINUX INSTALLATION STEPS

The Linux installation steps are listed below in full detail :-

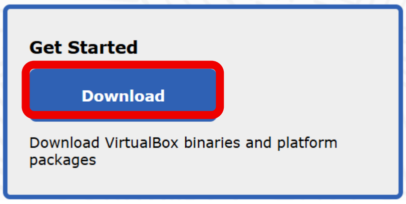
STEP-1 :- INSTALLING THE UBUNTU LASTEST VERSION

Visit the official site https://ubuntu.com/download/desktop .Click on the "Download for free" button. Now download the "**Ubuntu 24.04.3 LTS**" version of the ubuntu by clicking on the download button. Then ubuntu will be installed in your system.



STEP-2 :- INSTALLING THE ORACLE VIRTUAL BOX

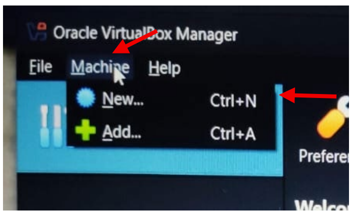
Visit the official site https://www.virtualbox.org/. Click on the "Download" button. Now, click on the "windows host" if you are working on windows otherwise select the option according to your system. Then oracle virtual box will be installed in your system.

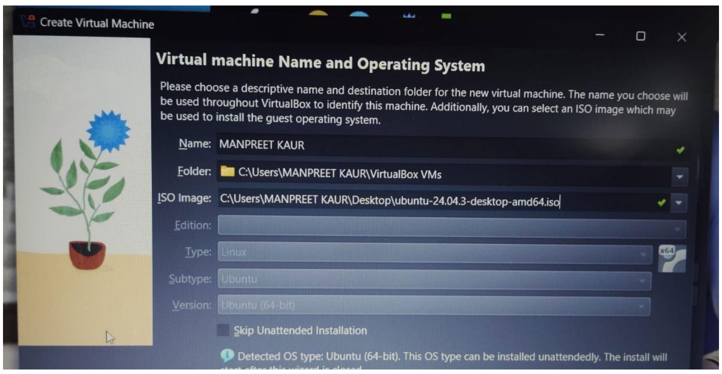


STEP-3 :- SETTING UP THE VIRTUAL BOX TO RUN UBUNTU IN IT

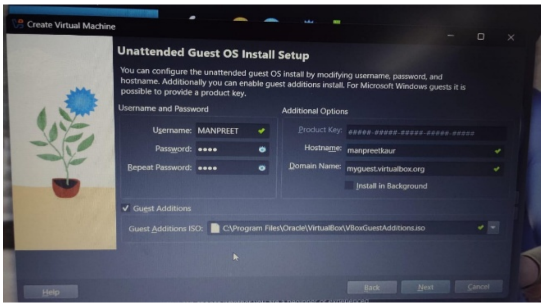
Now open the oracle virtual box and just go with the default options and click next and next and after this install this.

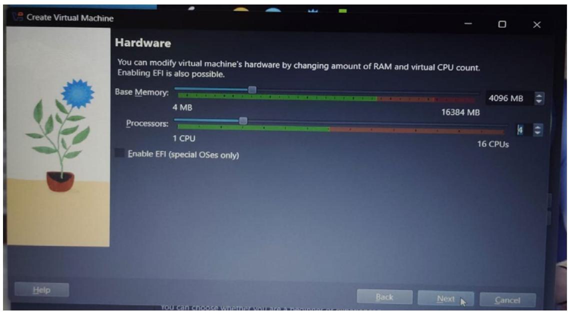
After all this follow the following steps:-

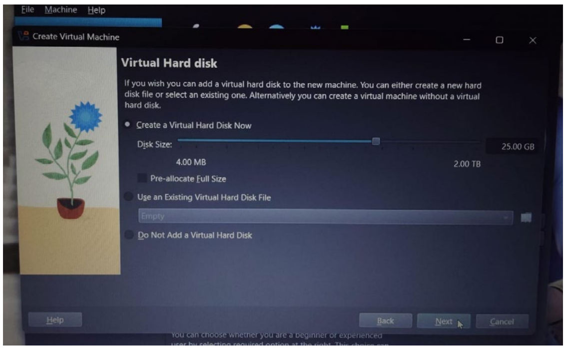




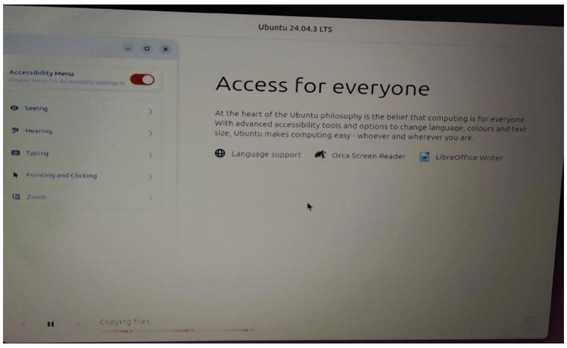
As you can see in the images that , when you open the virtual box you have to click on the "Machine" option. Then on the drop down menu appeared click on the "New" option. After all this you have to do the settings on the virtual machine that you are going to create. Fill the boxes as shown in the images and set your password. After that you have to select some hardware options such as giving the space to the "Base Memory" and the "CPU". This space is given according to the specifications of the system. I have given the maximum green space to my machine. After this you have to select the virtual hard disk options according to your choice, Here I have selected the option "Create a virtual hard disk now" having the space 25 GB. After that a window will open and all the installation process will start.





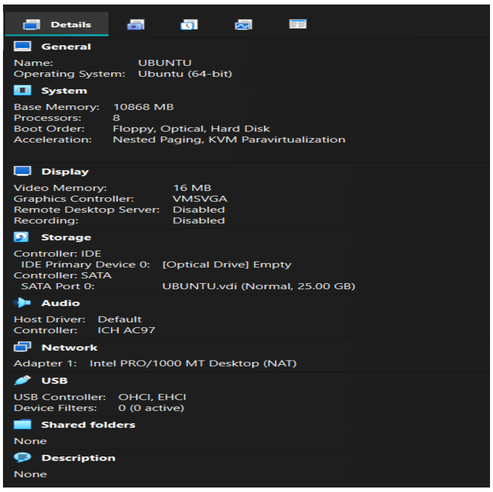


Now you are ready to work on Linux by using the Oracle Virtual Box, just open the virtual machine by clicking on the "Start" button and enter your password. After this open the "Terminal" in the virtual machine that you have created. Now you can practice all the Linux commands here easily.



These were all the installation steps ubuntu. For you can also refer the video given here for the installation:-

https://youtu.be/Hva8lsV2nTk?si=TEQ2FOWGN4QURJ_u



THESE ARE THE SPECIFICATIONS THAT I HAVE GIVEN TO MY ORACLE VIRTUAL BOX.

SHELL COMMAND IMPLEMENTATION AND DOCUMENTATION

[A] FILE NAVIGATION COMMANDS

These commands are used to navigate through the file in the system some of these commands such as pwd, ls, cd or tree etc. are listed in this document.

[B] FILE AND DIRECTORY MANAGEMENT COMMANDS

These Linux commands are used in directory and file managements such as creating then, removing them etc.. some of these commands such as mkdir, rmdir, rm, cp, mv, cat, touch or clear etc. are listed in this document.

[C] PERMISSIONS MANAGEMENT COMMANDS

Linux permission management primarily involves controlling who can read, write, and execute files and directories. Some of the commands such as chown and chmod are listed in this document.

[D] PROCESS MONITORING COMMANDS

Linux provides several commands for process monitoring, each offering different levels of detail and real-time capabilities. Some commands such as ps, top, kill etc. are listed in this document.

[E] NETWORKING TOOLS COMMANDS

Linux provides a comprehensive set of command-line networking tools for configuring, managing, and troubleshooting network connections. Some commands such as ping, ifconfig, netstat etc. are listed in this document.

(1) PWD COMMAND

DESCRIPTION :-

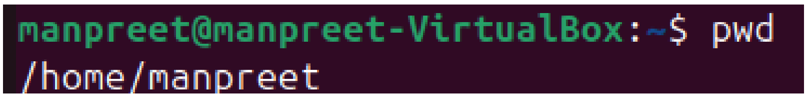
This command prints the current working directory, that means it tells you that where you are exactly in the file system hierarchy. The "pwd" command in Linux stands for "Print Current Working directory".

SYNTAX :- **pwd**

WHY AND WHEN :-

The pwd command in Linux is used to display the current working directorys full path. It is helpful to know your exact location in the file system, especially when navigating or scripting.

IMPLEMENTATION :-



(2) LS COMMAND

DESCRIPTION :-

The “ls” command in Linux is used to list the contents of the directories. When we execute this command without any arguments, it displays the files and subdirectories within the current working directory in alphabetical order.

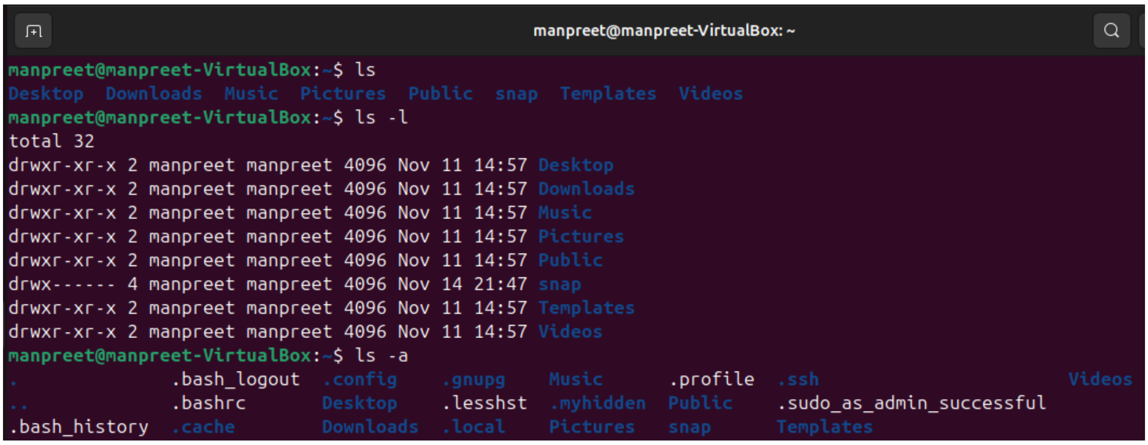
ls	List all the files / folders
ls -l	To get a detailed view of files / folders
ls -a	To show all the hidden files

SYNTAX :- ls [options] [file/directory]

WHY AND WHEN :-

The ls command in Linux is used to list the files and directories in the current or specified directory. It helps users quickly view what files are present for navigation or management. It’s commonly used to check directory contents efficiently in the terminal.

IMPLEMENTATION :-



(3) CD COMMAND

DESCRIPTION :-

The “cd” command in Linux is used to navigate the file system. It allows the user to change the current working directory in the terminal. Here “cd” stands for “change directory”.

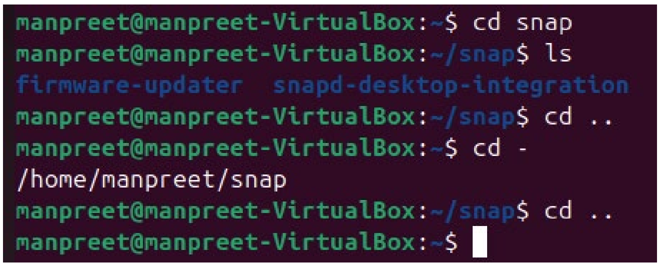
cd /path	Go to the specified folder
cd ..	Move up to the parent directory.
cd -	Switch to the previous directory.
cd	Jump to the home directory

SYNTAX :- **cd [options] [directory]**

WHY AND WHEN :-

The cd command is used to change the current working directory in Linux. It is commonly used whenever you need to navigate to a different directory to access or manage files efficiently in the command line environment

IMPLEMENTATION :-



(4) TREE COMMAND

DESCRIPTION :-

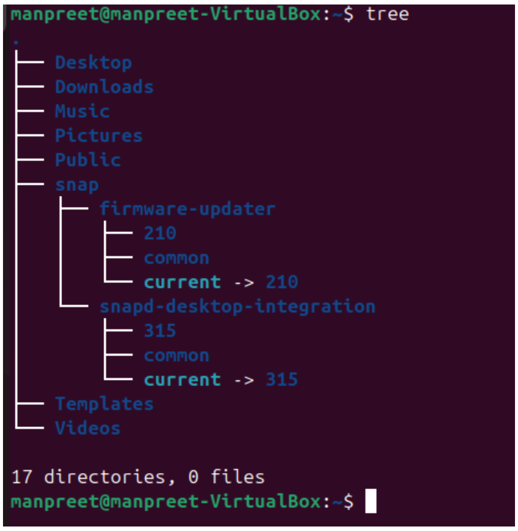
The tree command in Linux displays the directory structure in a clear tree-like format, showing files and subdirectories hierarchically.

SYNTAX :- **tree**

WHY AND WHEN :-

The tree command is used to quickly visualize the hierarchical structure of directories and files. It is helpful when you want to see the organization and nesting within a directory in a clear, easy-to-understand format.

IMPLEMENTATION :-



(5) MKDIR COMMAND

DESCRIPTION :-

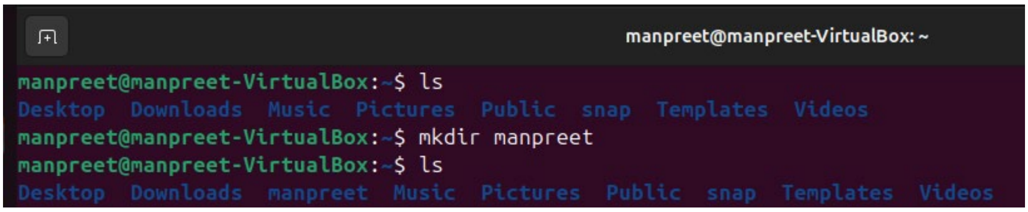
The command “mkdir” refers to “make directory”. This Linux command is used to create new directories in the file system.

SYNTAX :- **mkdir [directory_name]**

WHY AND WHEN :-

This command is used to create directories in the file systems. It is useful for organizing files by creating one or multiple directories quickly.

IMPLEMENTATION :-



(6) RMDIR COMMAND

DESCRIPTION :-

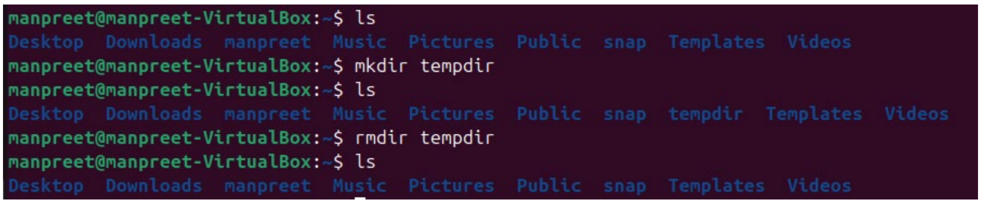
The command "rmdir" refers to "remove directory". This Linux command is used to remove an empty directory in the file system.

SYNTAX :- **rmdir [directory]**

WHY AND WHEN :-

This Linux command is used to remove the empty directories from the file system. It can only delete directories that do not contain any files or subdirectories, ensuring safe deletion without risking the loss of data inside. Commonly , this is used for cleaning up and managing the directory structures safely deleting unused or temporary empty folders.

IMPLEMENTATION :-



(7) CAT COMMAND

DESCRIPTION :-

The name "cat" command means "concatenate". It is used to display the content of files, concatenate multiple files, and redirect output to create or append to files.

SYNTAX :-

Viewing file content :- cat [filename]

Combining files :- cat [file1] [file2]

Creating or editing files with input redirection :- cat > [newfile]

WHY AND WHEN :-

The "cat" command in Linux is used to display the content of the file on the terminal or combine multiple files. It is commonly used to quickly view file contents or create new files by redirecting input. Its simplicity and versatility make it essential for file manipulation and quick content check.

IMPLEMENTATION :-

```
manpreet@manpreet-VirtualBox:~$ ls
Desktop  Downloads  manpreet  Music  Pictures  Public  snap  Templates  Videos
manpreet@manpreet-VirtualBox:~$ cat >file1.txt
hi,
i am manpreet kaur.
i am learning linux commands.
i am a b.tech cse student.
at k.r. mangalam university.
^Z
[1]+  Stopped                 cat > file1.txt
manpreet@manpreet-VirtualBox:~$ cat >file2.txt
this is the content of file2.txt i am using these files
for the implementation of the linux commands.^Z
[2]+  Stopped                 cat > file2.txt
```

```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
Desktop  Downloads  file1.txt  file2.txt  manpreet  Music  Pictures  Public  snap  Templates  touch.txt  Videos
manpreet@manpreet-VirtualBox:~$ cat file1.txt file2.txt > cat12.txt
```

```
manpreet@manpreet-VirtualBox:~$ ls
cat12.txt  Downloads  file2.txt  Music     Public  Templates  Videos
Desktop    file1.txt  manpreet   Pictures  snap    touch.txt
manpreet@manpreet-VirtualBox:~$ cat cat12.txt
hi,
i am manpreet kaur.
i am learning linux commands.
i am a b.tech cse student.
at k.r. mangalam university.
this is the content of file2.txt i am using these files
manpreet@manpreet-VirtualBox:~$
```

```
manpreet@manpreet-VirtualBox:~$ ls file1.txt
file1.txt
manpreet@manpreet-VirtualBox:~$ cat file1.txt
hi,
i am manpreet kaur.
i am learning linux commands.
i am a b.tech cse student.
at k.r. mangalam university.
manpreet@manpreet-VirtualBox:~$ cat file2.txt
this is the content of file2.txt i am using these files
manpreet@manpreet-VirtualBox:~$
```

(8) RM COMMAND

DESCRIPTION :-

The name “rm” stands for “remove files”. This command in Linux is used to permanently delete files. With options like -r for recursive deletion and -f for force deletion, it can remove directories and their contents safely and efficiently.

SYNTAX :- **rm [options] [file]**

WHY AND WHEN :-

This command is commonly used to remove unwanted files quickly from the command line. Use it with caution, as deleted files are not moved to a recycle bin and cannot be easily recovered.

IMPLEMENTATION :-

```
manpreet@manpreet-VirtualBox:~$ ls
Desktop  Downloads  file1.txt  file2.txt  manpreet  Music  Pictures  Public  snap  Templates  Videos
manpreet@manpreet-VirtualBox:~$ cd manpreet
manpreet@manpreet-VirtualBox:~/manpreet$ ls
file2.txt  file3.txt
manpreet@manpreet-VirtualBox:~/manpreet$ rm file2.txt
manpreet@manpreet-VirtualBox:~/manpreet$ ls
file3.txt
```

(9) CP COMMAND

DESCRIPTION :-

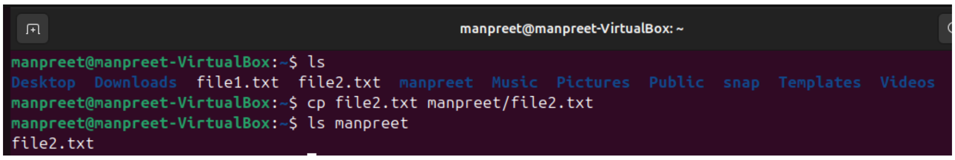
The name “cp” stands for “copy files”. This command is used to copy files or directories from one location to another.

SYNTAX :- **cp [options] source destination**

WHY AND WHEN :-

This command is mainly used when you want to duplicate files for backup, sharing or organizing data. This command is simple yet powerful for managing file copies efficiently within the Linux file system.

IMPLEMENTATION :-



(10) MV COMMAND

DESCRIPTION :-

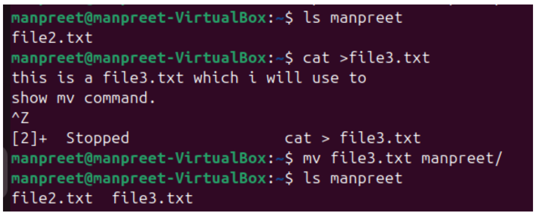
The word “mv” stands for “move files”. This command in Linux is used to move or rename files and directories.

SYNTAX :- **mv [options] source destination**

WHY AND WHEN :-

It moves files or directories from a source location to a destination. It can also rename files or directories by moving them to the same location with a different name. It is useful in organizing files, renaming files without copying/deleting.

IMPLEMENTATION :-



(11) CLEAR COMMAND

DESCRIPTION :-

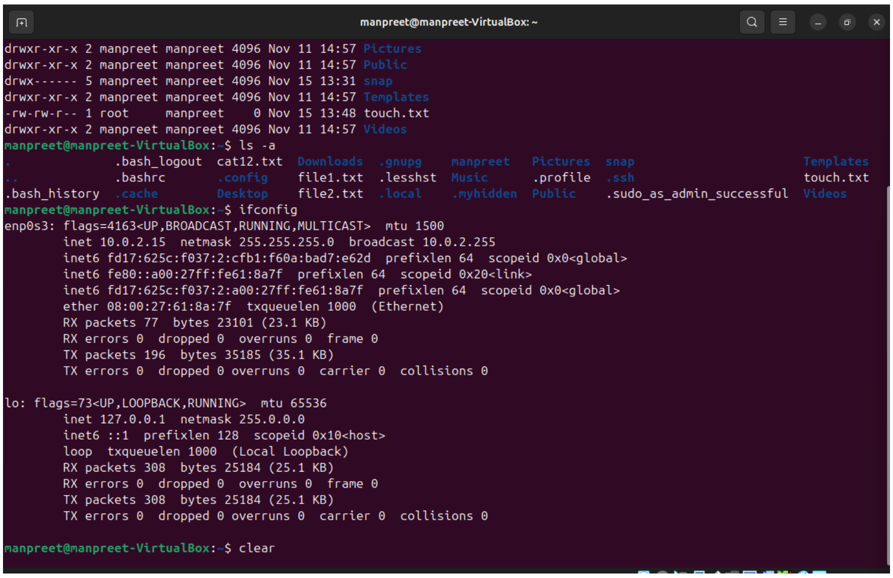
The “clear” command in Linux is used to clear the terminal screen, removing all previous commands and output from the visible area.

SYNTAX :- **clear**

WHY AND WHEN :-

It provides a clean workspace without closing the terminal session or deleting the command history. This command is useful for decluttering the terminal and improving readability during command-line work.

IMPLEMENTATION :-



(12) HEAD COMMAND

DESCRIPTION :-

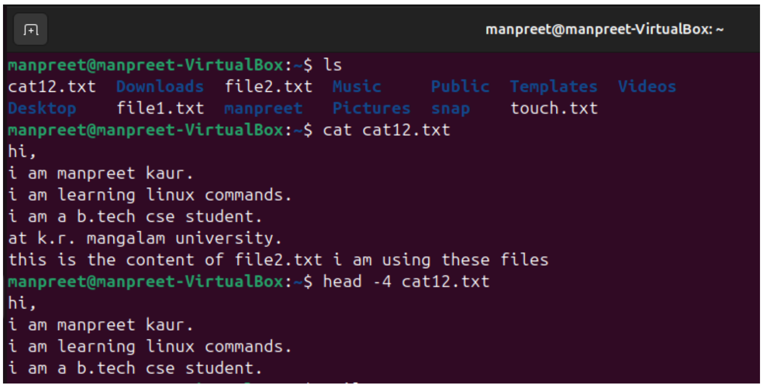
The “head” command in Linux is used to display the first few lines of a text file directly in the terminal. By default, it shows the first 10 lines but allows customization to show a specific number of lines or bytes.

SYNTAX :- **head [options] [files]**

WHY AND WHEN :-

It is commonly used to quickly preview the beginning of a log file, configuration file, or any large text file without opening the entire file. This makes “head” useful for quick inspection and troubleshooting in Linux environment.

IMPLEMENTATION :-



(13) TAIL COMMAND

DESCRIPTION :-

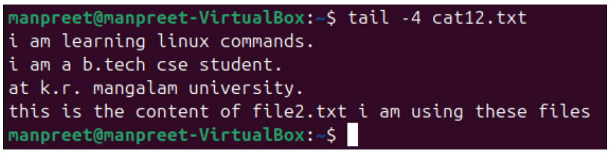
The “tail” command in Linux is used to display the last part of a file, typically the last 10 lines by default. It is commonly used to monitor log files or view recent updates added to files.

SYNTAX :- **tail [options] [files]**

WHY AND WHEN :-

The “tail” command helps quickly see the most recent content of a file and is essential for system administration and debugging tasks. This makes “tail” especially valuable for viewing recent parts of files or monitoring live updates.

IMPLEMENTATION :-



(14) CHMOD COMMAND

DESCRIPTION :-

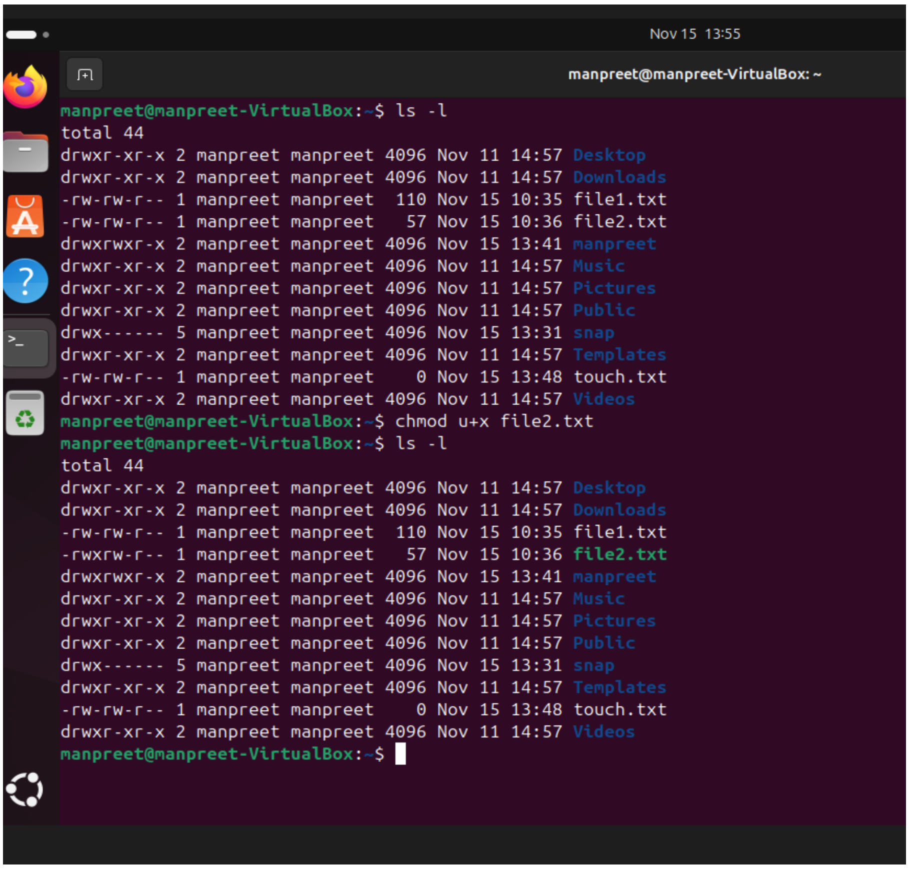
The “chmod” stands for “change mode”. This command in Linux is used to change the permissions (read, write and execute) of files and directories. These permissions determine who can access or modify the file or directory.

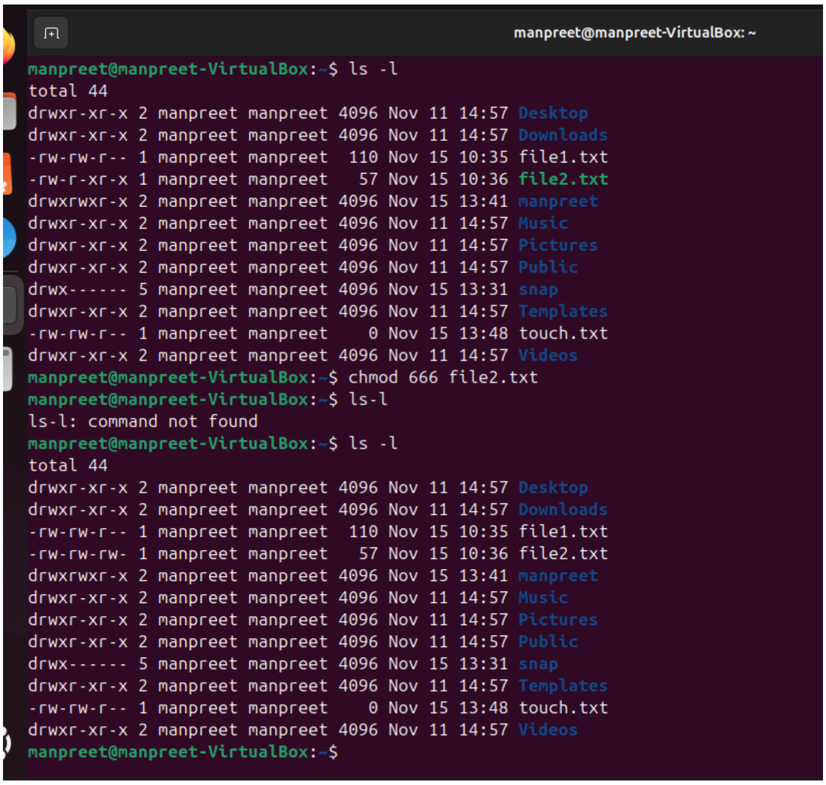
SYNTAX :- **chmod [options] mode file**

WHY AND WHEN :-

It allows setting permissions for the owner, group and others. Permissions can be set using symbolic (eg- u+rwx) or numeric (eg-755) modes. It helps control access and security for files and directories. This is essential for managing file security and controlling user access in Linux systems.

IMPLEMENTATION :-





```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls -l
total 44
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Desktop
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Downloads
-rw-rw-r-- 1 manpreet manpreet  110 Nov 15 10:35 file1.txt
-rw-r-xr-x 1 manpreet manpreet   57 Nov 15 10:36 file2.txt
drwxrwxr-x 2 manpreet manpreet 4096 Nov 15 13:41 manpreet
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Music
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Pictures
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Public
drwx------ 5 manpreet manpreet 4096 Nov 15 13:31 snap
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Templates
-rw-rw-r-- 1 manpreet manpreet    0 Nov 15 13:48 touch.txt
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Videos
manpreet@manpreet-VirtualBox:~$ chmod 666 file2.txt
manpreet@manpreet-VirtualBox:~$ ls-l
ls-l: command not found
manpreet@manpreet-VirtualBox:~$ ls -l
total 44
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Desktop
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Downloads
-rw-rw-r-- 1 manpreet manpreet  110 Nov 15 10:35 file1.txt
-rw-rw-rw- 1 manpreet manpreet   57 Nov 15 10:36 file2.txt
drwxrwxr-x 2 manpreet manpreet 4096 Nov 15 13:41 manpreet
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Music
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Pictures
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Public
drwx------ 5 manpreet manpreet 4096 Nov 15 13:31 snap
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Templates
-rw-rw-r-- 1 manpreet manpreet    0 Nov 15 13:48 touch.txt
drwxr-xr-x 2 manpreet manpreet 4096 Nov 11 14:57 Videos
manpreet@manpreet-VirtualBox:~$
```

(15) CHOWN COMMAND

DESCRIPTION :-

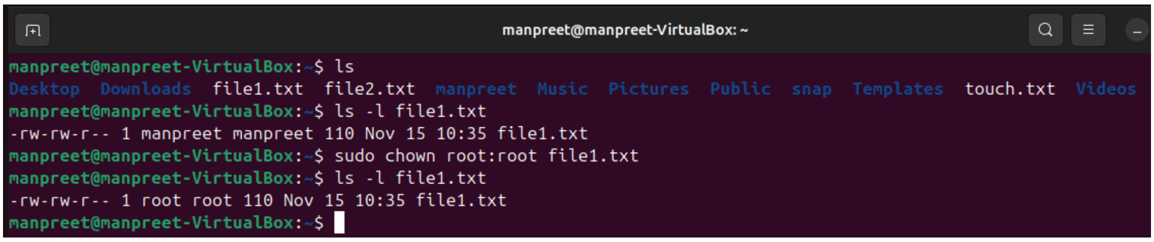
The "chown" stands for "change owner". This command is used to change the ownership of files or directories. It allows changing the user (owner) and optionally the group associated with a file or directory.

SYNTAX :- **chown [options] new_owner[:new_group] files**

WHY AND WHEN :-

It supports changing just the owner, just the group or both simultaneously. Commonly used to manage file permissions and access control on multi-user systems. Only the root user or the current file owner with proper permissions can change the ownership.

IMPLEMENTATION :-



```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
Desktop  Downloads  file1.txt  file2.txt  manpreet  Music  Pictures  Public  snap  Templates  touch.txt  Videos
manpreet@manpreet-VirtualBox:~$ ls -l file1.txt
-rw-rw-r-- 1 manpreet manpreet 110 Nov 15 10:35 file1.txt
manpreet@manpreet-VirtualBox:~$ sudo chown root:root file1.txt
manpreet@manpreet-VirtualBox:~$ ls -l file1.txt
-rw-rw-r-- 1 root root 110 Nov 15 10:35 file1.txt
manpreet@manpreet-VirtualBox:~$
```

(16) PS COMMAND

DESCRIPTION :-

The "ps" command in Linux is used to display information about active processed running on the system. It provides a snapshot view of current processed for the current user or all users depending on options.

SYNTAX :- **ps [options]**

WHY AND WHEN :-

It is useful for monitoring and managing system processes, reports processes for the current user or all users depending on options and can display detailed information or a simple list based on flags. “ps” is an essential command for system monitoring and process management in Linux.

IMPLEMENTATION :-

```
manpreet@manpreet-VirtualBox:~$ ps
    PID TTY          TIME CMD
   2895 pts/0    00:00:00 bash
   3018 pts/0    00:00:00 ps
```

(17) TOP COMMAND

DESCRIPTION :-

The “top” command in Linux provides a dynamic, real-time view of running processes and overall system resource usage. It displays system statistics including CPU and memory usage, process information and system load averages.

SYNTAX :- **top**

WHY AND WHEN :-

This command shows the active processes with PID, user, CPU, memory and runtime. It displays system uptime, number of users and load averages and many more things. This command is essential for monitoring system performance and manging processes in Linux.

IMPLEMENTATION :-

```
manpreet@manpreet-VirtualBox:~$ top
```

```
manpreet@manpreet-VirtualBox: ~
top - 14:58:18 up 12 min,  1 user,  load average: 0.46, 0.70, 0.72
Tasks: 235 total,   1 running, 234 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.1 us,  0.1 sy,  0.0 ni, 99.6 id,  0.2 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem :  10562.4 total,   7813.1 free,   1101.1 used,   1934.1 buff/cache
MiB Swap:   4096.0 total,   4096.0 free,      0.0 used.   9461.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2095 manpreet  20   0 5618920 399956 143672 S   2.3   3.7   1:30.69 gnome-shell
    537 systemd+  20   0   17560   7380   6612 S   0.3   0.1   0:01.51 systemd-oomd
    626 root      20   0       0      0      0 I   0.3   0.0   0:04.72 kworker/u32:6-kvfree_rcu_reclaim
   1876 manpreet  20   0   10800   6544   4624 S   0.3   0.1   0:04.22 dbus-daemon
   2888 manpreet  20   0  555336  53620  43084 S   0.3   0.5   0:07.57 gnome-terminal-
   3307 manpreet  20   0   14516   5844   3668 R   0.3   0.1   0:01.05 top
      1 root      20   0   23200  13936   9200 S   0.0   0.1   0:17.07 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.19 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
      9 root      20   0       0      0      0 I   0.0   0.0   0:00.07 kworker/0:0-events
     11 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     16 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     17 root      20   0       0      0      0 S   0.0   0.0   0:02.68 ksoftirqd/0
     18 root      20   0       0      0      0 I   0.0   0.0   0:06.31 rcu_preempt
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:05.88 rcu_exp_gp_kthread_worker
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.22 migration/0
```

(18) KILL COMMAND

DESCRIPTION :-

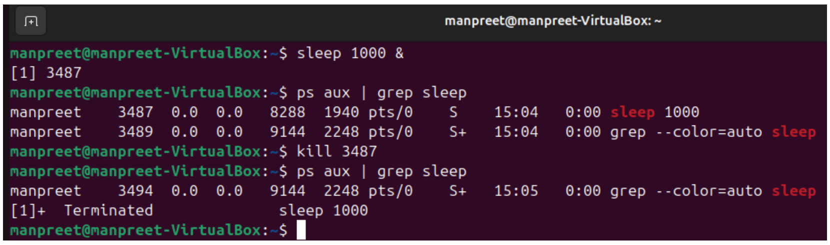
The “kill” command is used to send signals to processes, typically to terminate them. It sends a termination signal to process by its process ID (PID).

SYNTAX :- **kill [signal] <PID>**

WHY AND WHEN :-

This command is essential for managing and controlling the processes that run on the Linux system. Basically, it terminates the processes that we don't need.

IMPLEMENTATION :-



```
manpreet@manpreet-VirtualBox:~$ sleep 1000 &
[1] 3487
manpreet@manpreet-VirtualBox:~$ ps aux | grep sleep
manpreet    3487  0.0  0.0   8288  1940 pts/0    S    15:04   0:00 sleep 1000
manpreet    3489  0.0  0.0   9144  2248 pts/0    S+   15:04   0:00 grep --color=auto sleep
manpreet@manpreet-VirtualBox:~$ kill 3487
manpreet@manpreet-VirtualBox:~$ ps aux | grep sleep
manpreet    3494  0.0  0.0   9144  2248 pts/0    S+   15:05   0:00 grep --color=auto sleep
[1]+  Terminated              sleep 1000
manpreet@manpreet-VirtualBox:~$
```

(19) PING COMMAND

DESCRIPTION :-

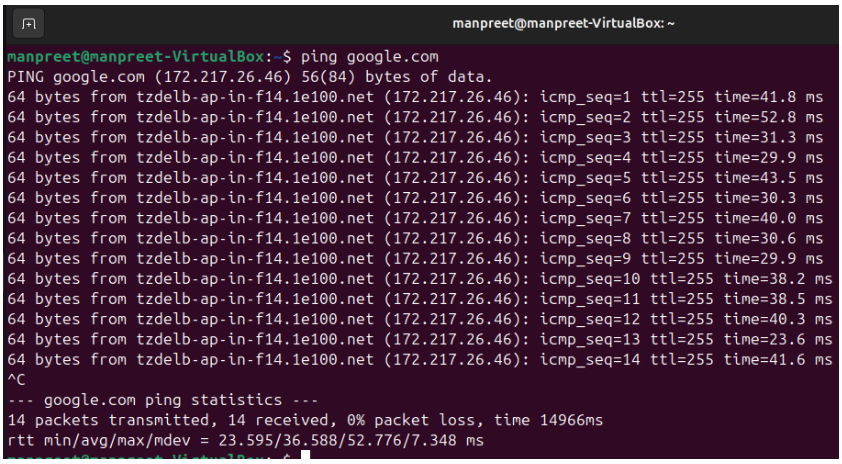
The "ping" command is a network utility command used to test the reachability of a host on an IP network and measure the round-trip time for messages sent from the originating host to a destination computer.

SYNTAX :- ping [options] destination

WHY AND WHEN :-

This command helps troubleshoot network connectivity by confirming whether a remote host is reachable and how long data packets take to travel there and back. It is one of the most commonly used commands for network diagnostics in Linux systems.

IMPLEMENTATION :-



```
manpreet@manpreet-VirtualBox:~$ ping google.com
PING google.com (172.217.26.46) 56(84) bytes of data.
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=1 ttl=255 time=41.8 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=2 ttl=255 time=52.8 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=3 ttl=255 time=31.3 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=4 ttl=255 time=29.9 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=5 ttl=255 time=43.5 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=6 ttl=255 time=30.3 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=7 ttl=255 time=40.0 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=8 ttl=255 time=30.6 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=9 ttl=255 time=29.9 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=10 ttl=255 time=38.2 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=11 ttl=255 time=38.5 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=12 ttl=255 time=40.3 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=13 ttl=255 time=23.6 ms
64 bytes from tzdelb-ap-in-f14.1e100.net (172.217.26.46): icmp_seq=14 ttl=255 time=41.6 ms
^C
--- google.com ping statistics ---
14 packets transmitted, 14 received, 0% packet loss, time 14966ms
rtt min/avg/max/mdev = 23.595/36.588/52.776/7.348 ms
```

(20) IFCONFIG COMMAND

DESCRIPTION :-

The "ifconfig" command in Linux is used to configure and display network interfaces. It can show information about network interfaces, assign IP addresses, activate or deactivate interfaces, and set various parameters.

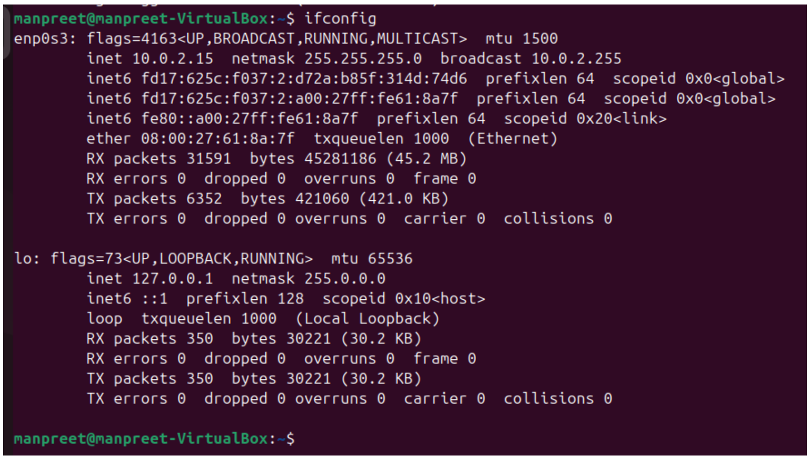
SYNTAX :- ifconfig [interface] [options]

WHY AND WHEN :-

"ifconfig" is a powerful network management tool that configures network interfaces, displays IP addresses, netmasks and hardware addresses and controls interface states.

This commands is considered legacy on some Linux distributions, replaced by the “ip’ command, but it is still widely used for network interface management.

IMPLEMENTATION :-



```
manpreet@manpreet-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:d72a:b85f:314d:74d6  prefixlen 64  scopeid 0x0<global>
        inet6 fd17:625c:f037:2:a00:27ff:fe61:8a7f  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::a00:27ff:fe61:8a7f  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:61:8a:7f  txqueuelen 1000  (Ethernet)
        RX packets 31591  bytes 45281186 (45.2 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 6352  bytes 421060 (421.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 350  bytes 30221 (30.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 350  bytes 30221 (30.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

manpreet@manpreet-VirtualBox:~$
```

(21) TOUCH COMMAND

DESCRIPTION :-

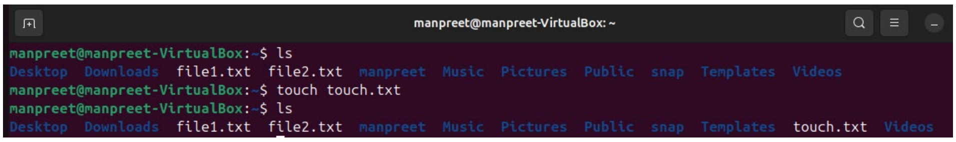
The ”touch” command in Linux is mainly used to create empty files or update the access and modification timestamps of existing files without changing their content.

SYNTAX :- **touch [options] file_name**

WHY AND WHEN :-

This command creates a new empty file if the file does not already exists, updates the access and modification times of an existing file to the current time and can also be used to set specific timestamps using options. Overall, this command is useful for scripting, file management and controlling file time attributes.

IMPLEMENTATION :-



```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
Desktop  Downloads  file1.txt  file2.txt  manpreet  Music  Pictures  Public  snap  Templates  Videos
manpreet@manpreet-VirtualBox:~$ touch touch.txt
manpreet@manpreet-VirtualBox:~$ ls
Desktop  Downloads  file1.txt  file2.txt  manpreet  Music  Pictures  Public  snap  Templates  touch.txt  Videos
```

(22) WHOAMI

DESCRIPTION :-

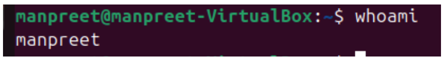
The “whoami” command in Linux displays the username of the current effective user-that is, the username under which the current process is running. This command can run without options also.

SYNTAX :- **whoami [options]**

WHY AND WHEN :-

This command simply prints the logged-in user’s name. it is useful for verifying the user identity, especially after switching users with commands like “sudo”. This command is often used in scripts to check or confirm the user running the scripts.

IMPLEMENTATION :-



```
manpreet@manpreet-VirtualBox:~$ whoami
manpreet
```

(23) NETSTAT COMMAND

DESCRIPTION :-

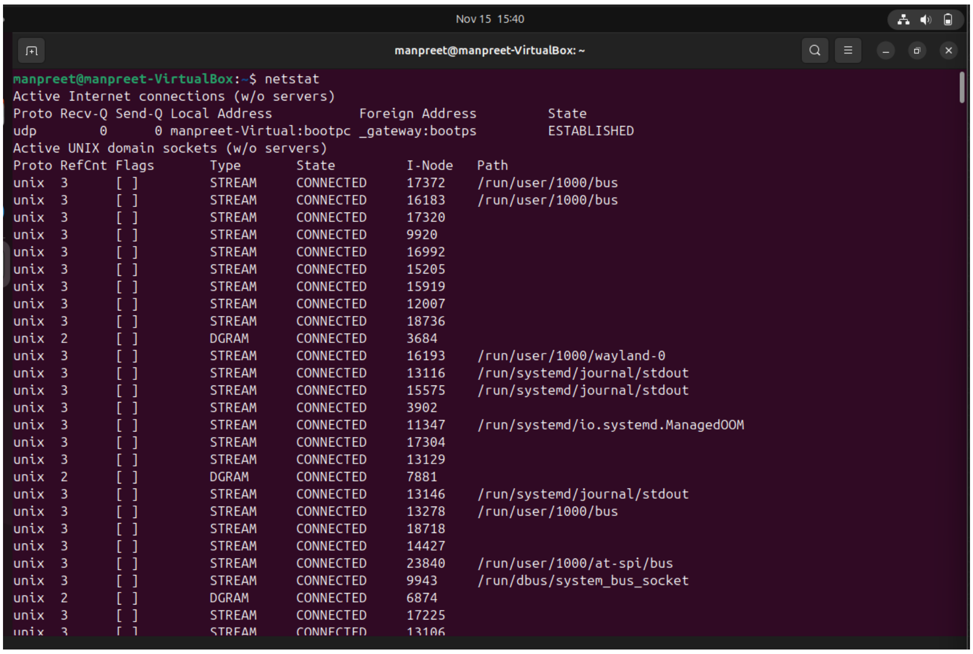
The “netstat” command in Linux is a network utility command used to display network connections, routing tables, interface statistics and multicast memberships etc.. It is commonly used for actively, showing which ports are open, active connections and which process is using a specific port.

SYNTAX :- **netstat [options]**

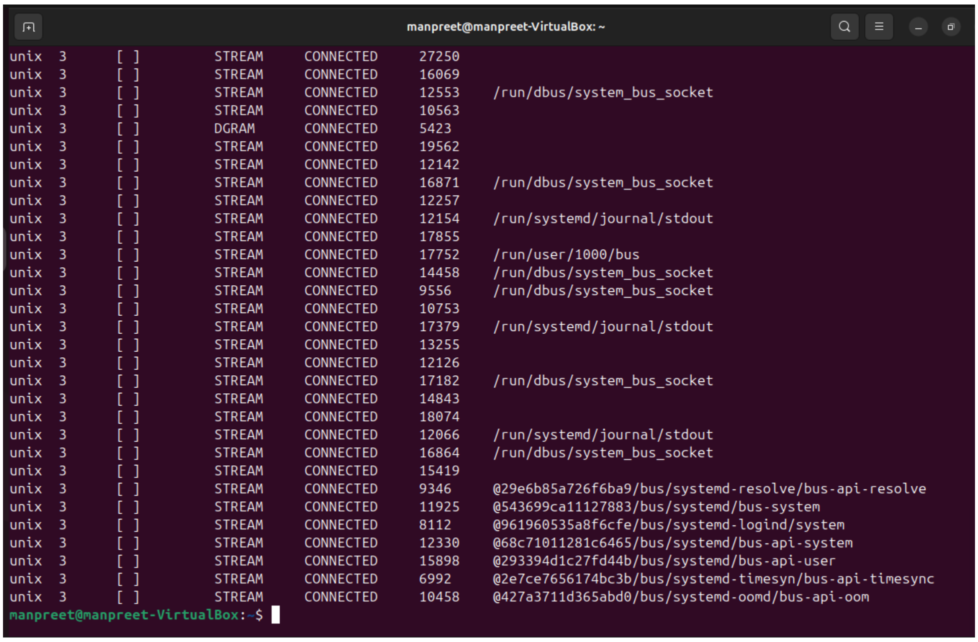
WHY AND WHEN :-

This command is essential for network administrators for monitoring and troubleshooting network issues. This command provides the detailed information on active TCP and UDP connections, displays listening ports, routing tables, and network interface statistics and helps diagnose network issues and monitor network traffic.

IMPLEMENTATION :-



```
manpreet@manpreet-VirtualBox:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 manpreet-Virtual:bootpc _gateway:bootps         ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     17372    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     16183    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     17320
unix  3      [ ]         STREAM     CONNECTED     9920
unix  3      [ ]         STREAM     CONNECTED     16992
unix  3      [ ]         STREAM     CONNECTED     15205
unix  3      [ ]         STREAM     CONNECTED     15919
unix  3      [ ]         STREAM     CONNECTED     12007
unix  3      [ ]         STREAM     CONNECTED     18736
unix  2      [ ]         DGRAM      CONNECTED     3684
unix  3      [ ]         STREAM     CONNECTED     16193    /run/user/1000/wayland-0
unix  3      [ ]         STREAM     CONNECTED     13116    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     15575    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     3902
unix  3      [ ]         STREAM     CONNECTED     11347    /run/systemd/io.systemd.ManagedOOM
unix  3      [ ]         STREAM     CONNECTED     17304
unix  3      [ ]         STREAM     CONNECTED     13129
unix  2      [ ]         DGRAM      CONNECTED     7881
unix  3      [ ]         STREAM     CONNECTED     13146    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     13278    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     18718
unix  3      [ ]         STREAM     CONNECTED     14427
unix  3      [ ]         STREAM     CONNECTED     23840    /run/user/1000/at-spi/bus
unix  3      [ ]         STREAM     CONNECTED     9943     /run/dbus/system_bus_socket
unix  2      [ ]         DGRAM      CONNECTED     6874
unix  3      [ ]         STREAM     CONNECTED     17225
unix  3      [ ]         STREAM     CONNECTED     13106
```



```
unix  3      [ ]         STREAM     CONNECTED     27250
unix  3      [ ]         STREAM     CONNECTED     16069
unix  3      [ ]         STREAM     CONNECTED     12553    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     10563
unix  3      [ ]         DGRAM      CONNECTED     5423
unix  3      [ ]         STREAM     CONNECTED     19562
unix  3      [ ]         STREAM     CONNECTED     12142
unix  3      [ ]         STREAM     CONNECTED     16871    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     12257
unix  3      [ ]         STREAM     CONNECTED     12154    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     17855
unix  3      [ ]         STREAM     CONNECTED     17752    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     14458    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     9556     /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     10753
unix  3      [ ]         STREAM     CONNECTED     17379    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     13255
unix  3      [ ]         STREAM     CONNECTED     12126
unix  3      [ ]         STREAM     CONNECTED     17182    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     14843
unix  3      [ ]         STREAM     CONNECTED     18074
unix  3      [ ]         STREAM     CONNECTED     12066    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     16864    /run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     15419
unix  3      [ ]         STREAM     CONNECTED     9346     @29e6b85a726f6ba9/bus/systemd-resolve/bus-api-resolve
unix  3      [ ]         STREAM     CONNECTED     11925    @543699ca11127883/bus/systemd/bus-system
unix  3      [ ]         STREAM     CONNECTED     8112     @961960535a8f6cfe/bus/systemd-logind/system
unix  3      [ ]         STREAM     CONNECTED     12330    @68c71011281c6465/bus/systemd/bus-api-system
unix  3      [ ]         STREAM     CONNECTED     15898    @293394d1c27fd44b/bus/systemd/bus-api-user
unix  3      [ ]         STREAM     CONNECTED     6992     @2e7ce7656174bc3b/bus/systemd-timesyn/bus-api-timesync
unix  3      [ ]         STREAM     CONNECTED     10458    @427a3711d365abd0/bus/systemd-oomd/bus-api-oom
manpreet@manpreet-VirtualBox:~$
```

SHELL SCRIPT DEVELOPMENT

BACKUP A DIRECTORY

A script that copies a specified directory to a backup folder with a timestamp.

The output of the script along with the script itself is shown in the snapshots below :-

```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
backupdir  Desktop    file1.txt  manpreet  Pictures  snap       touch.txt
cat12.txt  Downloads  file2.txt  Music     Public    Templates  Videos
manpreet@manpreet-VirtualBox:~$ nano script1.sh
manpreet@manpreet-VirtualBox:~$ ls
backupdir  Desktop    file1.txt  manpreet  Pictures  script1.sh  Templates  Videos
cat12.txt  Downloads  file2.txt  Music     Public    snap        touch.txt
manpreet@manpreet-VirtualBox:~$ chmod +x script1.sh
```

```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
backupdir  Desktop  Downloads  file2.txt  Music     Public    snap        touch.txt
cat12.txt  dir2     file1.txt  manpreet   Pictures  script1.sh  Templates  Videos
manpreet@manpreet-VirtualBox:~$ mkdir backup
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  Templates  Videos
backupdir  Desktop    Downloads  file2.txt  Music     Public    snap        touch.txt
manpreet@manpreet-VirtualBox:~$ ./script1.sh manpreet backup
Backup of 'manpreet' completed at 'backup/backup_20251116131018'
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  Templates  Videos
backupdir  Desktop    Downloads  file2.txt  Music     Public    snap        touch.txt
manpreet@manpreet-VirtualBox:~$ ls backup
backup_20251116131018
manpreet@manpreet-VirtualBox:~$ ls backup/backup_20251116131018
file3.txt
manpreet@manpreet-VirtualBox:~$ cat backup/backup_20251116131018/file3.txt
this is a file3.txt which i will use to
show mv command.
manpreet@manpreet-VirtualBox:~$
```

```
manpreet@manpreet-VirtualBox:~$ cat script1.sh
#!/bin/bash
# Backup a specified directory to a backup folder with a timestamp
# Author: Manpreet Kaur
# Date: 15 november 2025

# check if both arguments are given
if [ $# -ne 2 ]; then
        echo "Usage: $0 <source_directory> <backup_directory>"
        exit 1
fi

# Assinging the arguments to the variables
source=$1
backup=$2

# check if the source directory exists
if [ ! -d "$source" ]; then
        echo "Error: source directory '$source' does not exists."
        exit 1
fi

# creating the backup directory if it doesn't exist
if [ ! -d "$backup" ]; then
        mkdir -p "$backup"
fi

# Generating the timestamp for the backup folder name
timestamp=$(date +%Y%m%d%H%M%S)

# Copy the source directory to the backup directory with timestamp
cp -r "$source" "$backup/backup_$timestamp"
```

```
# Confirming the backup completion
echo "Backup of '$source' completed at '$backup/backup_$timestamp'"
```

CPU/ MEMORY MONITORING

A script that logs CPU and memory usage to a file at regular intervals.

The output of the script along with the script itself is shown in the snapshots below :-

```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  Templates  Videos
backupdir  Desktop    Downloads  file2.txt  Music     Public    snap        touch.txt
manpreet@manpreet-VirtualBox:~$ nano script2.sh
manpreet@manpreet-VirtualBox:~$ chmod +x script2.sh
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  snap       touch.txt
backupdir  Desktop    Downloads  file2.txt  Music     Public    script2.sh  Templates  Videos
manpreet@manpreet-VirtualBox:~$
```

```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  snap       touch.txt
backupdir  Desktop    Downloads  file2.txt  Music     Public    script2.sh  Templates  Videos
manpreet@manpreet-VirtualBox:~$ ./script2.sh 5 sys_monitoring.log
^C
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  snap                Templates  Videos
backupdir  Desktop    Downloads  file2.txt  Music     Public    script2.sh  sys_monitoring.log  touch.txt
manpreet@manpreet-VirtualBox:~$ cat sys_monitoring.log
Timestamp, CPU_Usage(%), Memory_Usage(%)
2025-11-16\ 14:38:26, 39.2, 13.7512
2025-11-16\ 14:38:31, 24, 13.7699
manpreet@manpreet-VirtualBox:~$
```

```
manpreet@manpreet-VirtualBox:~$ cat script2.sh
#!/bin/bash
# Log the CPU and memory usage to a file at regular intervals
# Author: Manpreet Kaur
# Date: 15 november 2025

# Check if both the arguments are given
if [ $# -ne 2 ]; then
        echo "Usage: $0 <interval_in_seconds> <output_file>"
        exit 1
fi

# Assigning the arguments to the variables
interval=$1
output=$2

# writing the header to the output file
echo "Timestamp, CPU_Usage(%), Memory_Usage(%)" > "$output"

# running the infinite loop to monitor and log CPU and memory usage
while true; do
        # Get the current timestamp
        timestamp=$(date +"%Y-%m-%d\ %H:%M:%S")

        # Get the CPU usage percentage
        cpu_usage=$(top -bn1 | grep "Cpu(s)" | awk '{print 100 - $8}')

        # Get the memory usage percentage
        mem_usage=$(free | grep Mem | awk '{print $3/$2 * 100.0}')

        # writing the data to the output file
        echo "$timestamp, $cpu_usage, $mem_usage" >> "$output"
```

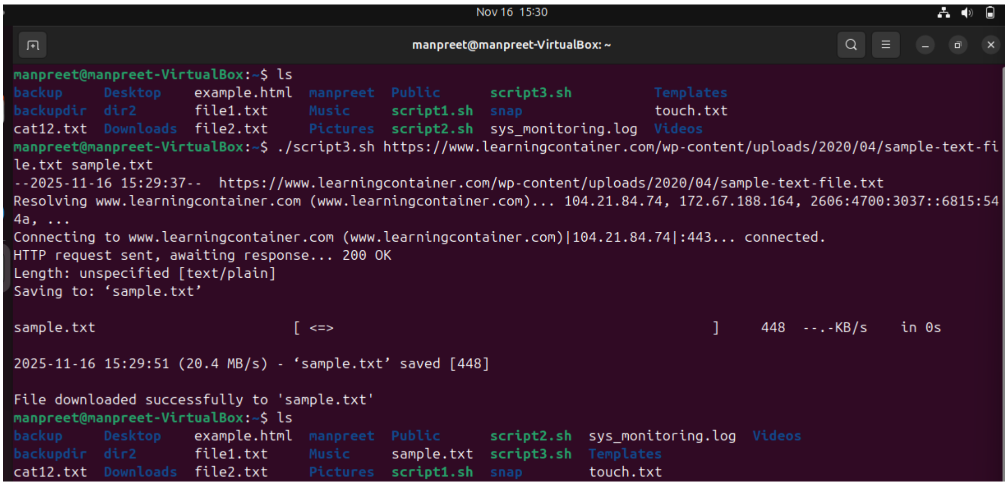
```
        # waiting for the specified interval
        sleep "$interval"
done
manpreet@manpreet-VirtualBox:~$
```

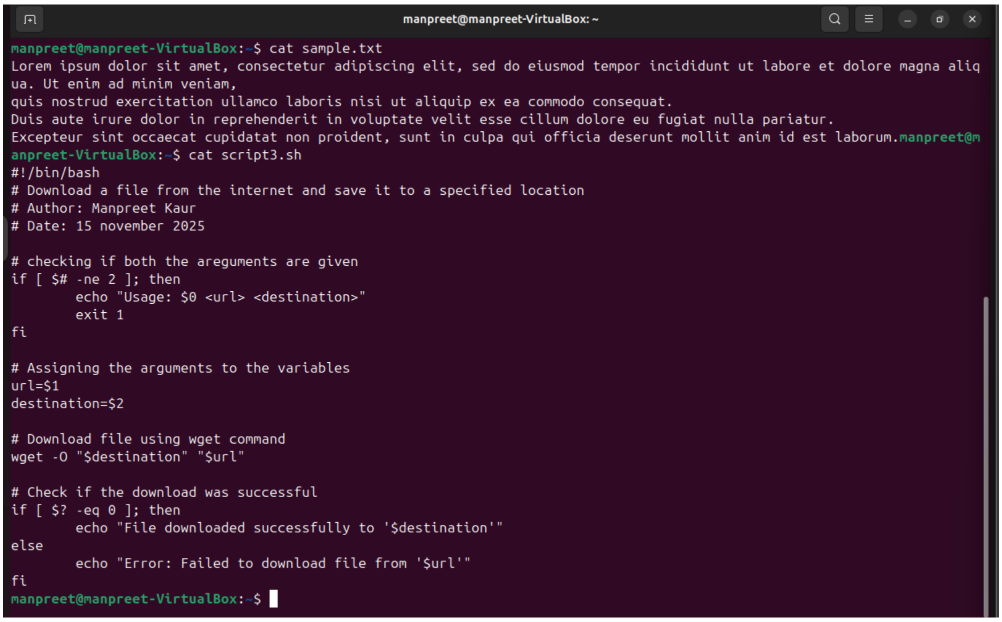
AUTOMATED DOWNLOAD TASK

A script that downloads a file from the internet using “wget” or “curl” and store it in a predefined directory.

The output of the script along with the script itself is shown in the snapshots below :-

```
manpreet@manpreet-VirtualBox: ~
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  snap                Templates  Videos
backupdir  Desktop    Downloads  file2.txt  Music     Public    script2.sh  sys_monitoring.log  touch.txt
manpreet@manpreet-VirtualBox:~$ nano script3.sh
manpreet@manpreet-VirtualBox:~$ chmod +x script3.sh
manpreet@manpreet-VirtualBox:~$ ls
backup     cat12.txt  dir2       file1.txt  manpreet  Pictures  script1.sh  script3.sh  sys_monitoring.log  touch.txt
backupdir  Desktop    Downloads  file2.txt  Music     Public    script2.sh  snap        Templates           Videos
```





GITHUB REPOSITORY LINK

https://github.com/manpreetkaur292006-design/LINUX-COMMANDS-AND-SCRIPT

REFLECTION

While doing this assignment, I got practical experience in using and combining Linux commands inside scripts, which made many jobs easier and faster. I learned how to check arguments, create and manage directories, and handle errors in my scripts. These skills will be very useful for my future studies and work, because scripting is important for any computer professional. I also gained confidence in using the terminal and writing scripts, which will help me with more advanced topics later on.

THANK YOU